

Representation Convergence — Results Summary

Every experiment's measured outcome in one place. 34 verdicts | 21 confirmed | 7 falsified | 4 inconclusive | 2 not meaningful. Full method in Final_Project_Report.pdf (84 pp).

The findings at a glance

#	finding	headline evidence
1	Convergence is linear	MLP = ridge to 4 decimals (0.9183 = 0.9183), two independent runs
2	Related, not rigid	Procrustes R ² 0.038 vs ridge 0.592; 69× singular-value spread
3	Deployable	95% of ceiling at 0.79 MB; verification AUC 0.999
4	Cross-modal without joint training	DINOv2 → bge 40.9–58.3% of ceiling, up to 358× chance
5	Alignment scales with capacity (bounded by E.16)	+15.3 pts/decade, six monotone measures; both ladders FALL beyond this range (E.16 FALSIFIED)
6	Isotropy, not similarity-training	whitening → 102.6% / 101.4% of a contrastive encoder at two scales
7	Correspondence is self-identifying	Gromov-Wasserstein 39.7% exact = 595× chance, zero pairs
8	One shared space, used	hub transfer 92.9–96.7% of native on unseen encoders incl. cross-lineage
9	The mechanism predicts	pre-registered encoder ladder, rank correlation +1.00

Meta-finding, measured twice: isotropy helps READING a correspondence (C.11) and hurts FINDING one (C.12). Same geometric property, opposite prescriptions.

Experiment A — GPT-2 vs Pythia-160M

metric	fail	pass	measured
CKA diagonal mean (trained)	<0.15	>0.35	0.424
CKA mean, random control	>0.20	<0.15	0.067
trained / random ratio	<2×	≥3×	6.3×
stitching R ² , peak L11	<0.50	>0.70	0.797
R ² mean across 13 layers	—	—	0.737
depth-slope signature	both flat	opposite	0.59→0.80 vs 0.38→0.09

Verdict: PASS. Run 1 (1,646 samples) failed on sample starvation; run 2 (10,000) passed.

Experiment B — MobileCLIP-S1 → SigLIP 2

quantity	value
Ridge adapter: held-out R ² / mean cosine	0.592 / 0.900
Procrustes (rotation only): R ²	0.038
MLP adapter: best cosine	0.903 (+0.003 → relationship is linear)
Retrieval R@1/R@5/R@10 (ceiling)	0.630 / 0.885 / 0.944
Retrieval with adapter (% ceiling)	88.4% / 95.3% / 97.2%
Raw cross-space baseline	0.001 / 0.005 / 0.010 (exact chance)
Deployed artifact	0.79 MB fp16 Core ML, iOS 16+

Series E, F, C, G — how far the claim goes

experiment	result
E1 capacity sweep	s/b/l R@1 40.9/53.5/58.3% of ceiling; +15.3 pts/decade, R ² 0.949
E1.1 GPT-2 collapse	R ² retains 71%, retrieval collapses; cosine 1.000, R@1 0.160
E1.2 isotropy rescue (C.11)	whitening R@1 0.130→0.478 = 102.6%; eff.rank 6.5→490.8
E1.3 encoder ladder	P1 rank-corr +1.00 (confirmed); P2 partial — necessary, not sufficient
E.16 scale extension	DINOv2-giant ρ 0.339 < large's 0.418: P2 FALSIFIED. Qwen-4B ρ 0.307 < bge's 0.320: P3 FALSIFIED. Both ladders rise then fall. Not monotonic.
F1/F2 capacity test	MLP = ridge: 0.9183 = 0.9183, gain +0.0000, two runs
F5 data test	+0.0244, 90%-gate NOT met at 89.0%
C / vec2vec	39.7% exact = 595× chance; whitening destroys it (→0.1%)

Series G — the shared hub

experiment	result
G1 seven-space hub	R@1 0.451 at 512-d; 42/42 pairs at 103.6% of bespoke
G1 read/write	GPT-2 source 95.1%, target 24.8% — readable, not writable
G4 cross-lineage	SigLIP 2: 94.2% of native; ConvNeXt: 96.7% — above the band
G5 what explains it	objective +53× > lineage +44× > modality +29×
G5 CKNNA	21/21 higher locally than globally; agreement is LOCAL
G6 manifold	intrinsic dim 11–20 in 768–2048-d spaces (<3%); CCA 6/64 shared
G7 local structure	neutral on healthy spaces; CI [−0.038, +0.024], MDE 0.034
G11–G16 width cliff	7 accounts falsified; 8th (RANK WALL) confirmed to single direction; σ 0.005 vs 10.79; truncation or $\alpha=1e3$ repairs it
G10 hub composition	text-member identity moves transfer 6.1pt; target-in-hub +3.0pt

G8–G10 — fractal probe (post-defense)

experiment	result
G8 training-free suite	4 metrics × 8 encoders: raw shape- ρ 0.288 (healthy-only 0.329); whitened 0.062. bge–SBERT 0.771. GPT-2 outlier.
G9 fractal models vs roster	P1 confirmed 8/8; P2 confirmed 4/4; P3 INCONCLUSIVE; P5-cnn falsified 0/4 (raw), P5w confirmed 4/4 (whitened). R ≈ 0.09 (CNN) / 0.21 (ViT).
G9 preprocessing control	rand_cnn_fnorm: norm ~10% of signal, training ~90%. P2-cnn survives.
G10 hub recoverability	fractal CNN 0.014, ViT 0.020 R@1 — AT FLOOR (= random twin). H1 FALSIFIED. H3 NOT MEANINGFUL (floor guard). H2 CONFIRMED (ImageNet 0.258/0.204 clears floor).

The fractal story: visible geometry exists (R 9–21%, P2 confirmed) but does NOT transfer through the frozen hub (H1 falsified). Natural-world data is load-bearing.

Falsification ledger — 11 explanations tested and falsified

#	explanation	verdict	where
1	collapse depresses shape corr.	FALSIFIED	C.11.2
2	hub discards distinguishing dirs	FALSIFIED	C.13
3	reach predicts writability	FALSIFIED	C.13.4
4	more encoders improve transfer	FALSIFIED	C.13
5	ratio-to-chance decline = locality	FALSIFIED (own)	C.13.9
6	cliff is local-structure loss	FALSIFIED	C.13.11
7	cliff is spectral amplification	FALSIFIED (own)	C.13.11
8	four cliff statistics	ALL FALSIFIED	E.3
9	cliff at head target's dimension	FALSIFIED	E.3
10	cliff = ill-conditioned accumulation	FALSIFIED	E.8
11	fractal geometry recoverable through hub	FALSIFIED	G10, F.4

Cached vectors, dimensions and splits

space	cache file	rows	dim	norm	used in
img_small	e1_img_ckpt_dinov2-small_cls+patch	9,533	768	54.70	hub, all pair tests
img_base	e1_img_ckpt_dinov2-base_cls+patch	18,066	1,536	52.91	hub; first 9,533
img_large	e1_img_ckpt_dinov2-large_cls+patch	23,755	2,048	50.83	hub; first 9,533
txt_bge	crossmodal_pairs.npz [txt]	18,066	1,024	0.89	hub member + head target
txt_gpt2	crossmodal_pairs_gpt2.npz [txt]	18,066	768	207.78	hub; C.11 collapse case
txt_bert	e13_txt_bert.npz	18,066	768	9.15	hub; encoder ladder
txt_sbert	e13_txt_sbert.npz	18,066	768	2.43	hub; best target (101%)
SigLIP 2	e1_img_ckpt_siglip2-base...	9,533	768	11.38	HELD OUT — cross-lineage
ConvNeXt	e1_img_ckpt_convnext-base...	9,533	1,024	16.70	HELD OUT — cross-arch
DINOv2-giant	cache_img_giant.npz	9,533	3,072	—	scale extension (E.16)
Qwen-0.6B	cache_txt_qwen06.npz	9,533	1,024	—	scale extension (E.16)
Qwen-4B	cache_txt_qwen4.npz	9,533	2,560	—	scale extension (E.16)
frac_cnn	g9_frac_cnn_9533.npz	9,533	2,048	17.60	fractal probe (G9/G10)
frac_vit	g9_frac_vit_9533.npz	9,533	384	35.58	fractal probe (G9/G10)
rand_cnn	g9_rand_cnn_9533.npz	9,533	2,048	1155	fractal control
rand_cnn_fnorm	g9_rand_cnn_fnorm_9533.npz	9,533	2,048	—	preprocessing control
rand_vit	g9_rand_vit_9533.npz	9,533	384	15.38	fractal control
nat_cnn	g9_nat_cnn_9533.npz	9,533	2,048	29.30	ImageNet twin (CNN)
nat_vit	g9_nat_vit_9533.npz	9,533	384	27.02	ImageNet twin (ViT)
pixels	g9_pixels_9533.npz	9,533	3,072	13.65	pixel floor
Experiment B	pairs.npz	4,000	512/768	1.00	adapter, B2–B10, C1
distillation	distill_corpus.npz	39,999	768/512	1.00	F-series

Splits, pairs and counts

• Hub protocol (G1, G4, G0-exact, G10): 9,533 shared images, 8,533 train / 1,000 eval, seed 0. • Experiment B: 4,000 COCO, 3,000/1,000, seed 0, one caption per image. • E and G series: five COCO captions per image, averaged. • Distillation (F1–F5): 39,999 rows, 37,999/2,000. • Pair counts: 7 spaces → 21 unordered / 42 ordered. • Rows/dim: every fit 5.9–9.8, above floor of 5. • Deployed: 512×768 ridge, 0.79 MB Core ML fp16.

Project verdict ledger — 34 pre-registered predictions

verdict	count	includes
CONFIRMED	21	shared geometry, cross-modal/lineage transfer, hub 93–96%, fractal geometry above random (P1/P2), encoder ladder
FALSIFIED	7	fractal geometry recoverable through hub (no); naive scaling beyond 304M (both ladders); MLP > ridge
INCONCLUSIVE	4	image-vs-text reality-dependence (P3 CIs straddle zero, both architectures, raw and whitened)
NOT MEANINGFUL	2	H3 CNN/ViT %-of-native blocked by 0.050 floor guard; both transfer and native at floor

Eleven falsified explanations, two the project's own. Nine findings, each held-out and each quoted against a control. The fractal probe (post-defense) confirmed that fractal training creates real shared geometry but does not produce a training-specific transfer advantage — natural-world data is load-bearing, and visible geometry alone is not enough for recoverable semantic alignment.